

Lefschetz Thimbles for Simple Zero-dimensional Systems

(Project Report)



Department of Physical Sciences
Indian Institute of Science Education and Research (IISER)
Mohali

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Author - Roshan S. Kaundinya Supervisor - Dr. Anosh Joseph

Abstract

In this report we outline some elementary numerical and analytical methods for finding Lefschetz thimbles and their duals. Most of the complex actions studied in this project describe fermionic and supersymmetric systems. We investigated the method of thimble cooling, which is a numerical method to acquire the thimble structures, in models including fermions. Straightforward analytical methods are employed to solve for actions defining supersymmetric quantum mechanics. We further highlight the difficulties with both these methods.

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Roshan S. Kaundinya
IISER Mohali
September 26, 2020

Contents

1	Introduction	5
2	Thimble Cooling	5
2.1	Thimble Cooling for $S(\phi) = \beta\phi^4 + \alpha\phi^2$	6
2.2	Thimble Cooling for $S(\sigma) = \frac{\sigma}{G} - \ln(p^2 + (\sigma + m)^2)$	7
3	Demerits of the Algorithm	10
4	Algorithm Description with Code Snippets	12
5	Supersymmetric Setup	18
6	Double-Well Potential	18
6.1	Real and Imaginary Parts	19
6.2	Critical Points	19
6.3	Parameterizations	19
6.4	Cases Considered	20
7	Scarf I Potential	20
7.1	Critical Points	21
7.2	Branch Points and Branch Cuts for $\ln \cos(\phi)$	22
7.3	Some Thimble and Region Analysis	22
7.4	Behavior of the thimbles under change of parameter A	23
8	Morse Potential	23
8.1	Critical Points	24
8.2	Real and Imaginary Parts of the Action	24
8.3	Analytic Expressions for Level Sets $\text{Im}(S[\phi]) = \text{Im}(S[\phi_{\pm}])$	24
8.4	Complexifying $B \rightarrow e^{i\theta}$	25
9	Closing Remarks	26

1 Introduction

This report is two fold. One part of it describes the working towards first implementing thimble cooling algorithm [4] to already analytically solved systems that can be found in the literature [2, 3]. The rest of the report looks at solving for thimbles and their duals analytically for a zero-dimensional supersymmetric (SUSY) action, for three different superpotentials [1, 5]. The later part of this work is still in progress. The main motivation for performing the thimble mode of analysis was to see if non-perturbative SUSY breaking leads to some significant change in the structure of thimbles and their duals. We are yet to decide upon a method to observe these effects, and this report just highlights the structure for these systems without accounting for SUSY breaking.

The thimble cooling part of this report analyzed a degree-4 polynomial action and a zero-dimensional Gross-Neveu (GN) like model. The GN like model's thimble analysis has been carried out and contours have been plotted. The initial phase of this project tried to reproduce these plots but due to numerical reasons we shifted to the thimble cooling method to achieve our goal.

In the thimble analysis of the zero-dimensional supersymmetric action the superpotentials used were double-well, Scarf I and Morse.

2 Thimble Cooling

To acquire thimble and anti-thimbles, we can look at contour plots for the following equation

$$\text{Im}S(x, y) = \text{Im}S(x_c, y_c), \quad (1)$$

where c denotes the critical point of the action

$$\left. \frac{\partial S}{\partial \phi} \right|_{z_c} = 0. \quad (2)$$

The thimble cooling (TC) algorithm focuses on minimizing the function

$$M_{z_c} = [\text{Im}S(z) - \text{Im}S(z_c)]^2, \quad (3)$$

where $z = x + iy$.

The minimization procedure results in the gradient equations whose solutions are curves that are orthogonal to the constant curves (thimbles, anti-thimbles, and the ghosts).

The gradient equations are

$$\dot{x} = -\frac{\partial M_{z_c}}{\partial x}, \quad (4)$$

$$\dot{y} = -\frac{\partial M_{z_c}}{\partial y}. \quad (5)$$

The above set of ordinary differential equations can be solved for on a grid of initial conditions. This grid could be uniform or random.

In the end, the set of points obtained could be filtered out and plotted. Later on, if the parametrizations are of interest, we could interpolate the data sets in the given interval and thus numerically evaluate the partition function in that domain of interpolation. This might work best with polynomial or periodic actions. In the subsections given below we will follow a typical format. The format starts from setting up the critical points of the action and after that finding analytical solutions of thimbles if solvable, and then followed by images of some interest. (See Ref. [4] for more details.)

2.1 Thimble Cooling for $S(\phi) = \beta\phi^4 + \alpha\phi^2$

Given the above action, we find the first derivative. It has the form

$$\frac{\partial S}{\partial \phi} = 4\beta\phi^3 + 2\alpha\phi. \quad (6)$$

Setting this to 0 gives us the critical points of interest

$$\phi = 0, \quad (7)$$

$$\phi = \pm i\sqrt{\frac{\alpha}{\beta}}. \quad (8)$$

Here $\alpha, \beta \in \mathbb{C}$. If we now set them to 1 for simplicity our critical points are 0 and $\pm i$. We will have to further test these points for the character of non-degeneracy. This is firstly done by calculating the Hessian of the complexified S . In some sense the calculations would amount to treating the effective problem in $\mathbb{R}^2$. The points are non-degenerate if the Hessian of S at z_c is non-zero.

We have

$$\text{Im}(S[\phi]) = xy(x^2 - y^2 + 1). \quad (9)$$

We see that $\text{Im}(S[z_c]) = 0$ for all the critical points. It is easy to see that $x = 0$, $y = 0$ and $y^2 - x^2 = 1$ are the contours. Also, by inspection we can argue that

$\text{Re}(S[x, y = 0]) \geq 0$. Therefore, $y = 0$ is the thimble since it is in the region of convergence.

In the language of the TC approach, for each critical point, we have a set of ODEs. The algorithm takes a uniform grid of initial conditions for a set of ODEs and solves for the trajectories. These trajectories would end up getting attracted to a thimble or anti-thimble. In our case we are left with just one set of ODEs since all critical points have a common contour equation

$$\dot{x} = -(6x^5y^2 + 2xy^6 + 2xy^2 + 8x^3y^2 - 4xy^4 - 8x^3y^4), \quad (10)$$

$$\dot{y} = -(6y^5x^2 + 2yx^6 + 2yx^2 - 8y^3x^2 + 4yx^4 - 8y^3x^4). \quad (11)$$

The analysis was also done for a system with critical points providing us with different set of ODEs. When we take $\alpha = 1 + i$ and $\beta = 1$, the critical points were found to be $\phi_{0,\pm} = 0, \pm\sqrt{-1-i}$. The imaginary part of the action for this set up has the form

$$\text{Im}(S[\phi]) = xy + xy(x^2 - y^2) + \frac{x^2 - y^2}{2}. \quad (12)$$

Hence the M's in TC approach would be

$$\begin{aligned} M_0 &= \left(xy + xy(x^2 - y^2) + \frac{x^2 - y^2}{2} \right)^2, \\ M_{+,-} &= \left(xy + \frac{1}{2} \right)^2 (1 + x^2 - y^2)^2. \end{aligned} \quad (13)$$

So the following example provides us with two sets of ODEs. These figures can be verified with plots from Ref. [2]. See Figs. 1 through 5 for all the plots.

Note: To evaluate the Hessian of $S(\mathbf{H}(S))$ we will need to write S in the form $u + iv$ where $u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$. We can think of $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, in some sense $S \doteq (u, v)$ and $\mathbf{H}(S) \doteq (\mathbf{H}(u), \mathbf{H}(v))$. Accounting for S to be Cauchy-Riemann differentiable we see that $\mathbf{H}(v) = \mathbf{H}(u) = f(u_{xx}, u_{yy}, v_{xx}, v_{yy})$.

2.2 Thimble Cooling for $S(\sigma) = \frac{\sigma}{G} - \ln(p^2 + (\sigma + m)^2)$

This zero-dimensional fermionic action

$$S(\sigma) = \frac{\sigma}{G} - \ln(p^2 + (\sigma + m)^2) \quad (14)$$

can be obtained from a Hubbard-Stratonovich transformation for a GN like model partition function of the form

$$Z_N(G, m) = \int d\bar{\psi} d\psi \exp \left[\sum_{a=1}^N \bar{\psi}_a (i\not{p} + m) \psi_a + \frac{G}{4N} \left(\sum_{a=1}^N \bar{\psi}_a \psi_a \right)^2 \right], \quad (15)$$

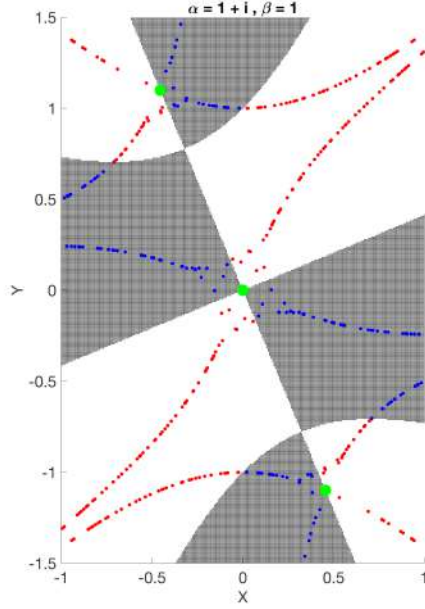


Figure 1: Shaded regions depict the regions of stability, $\text{Re}(S[\phi]) \geq 0$. The blue and red dots represent thimbles and anti-thimbles, respectively. The green blobs represent critical points. We used the parameters $\alpha = 1 + i$ and $\beta = 1$.

where $\not{p} = \sum_{i=1}^2 p_i \sigma_i$. Substituting $\sigma = \sum_{a=1}^N \bar{\psi}_a \psi_a$, and a subsequent Gaussian integral (Hubbard-Stratonovich transformation) is performed to reduce it to look like

$$Z_N(G, m) = \sqrt{\frac{N}{\pi G}} \int d\bar{\psi} d\psi d\sigma \times \exp \left(\sum_{a=1}^N \bar{\psi}_a (i\not{p} + m + \sigma) \psi_a - \frac{N}{G} \sigma^2 \right). \quad (16)$$

Integrating over the fermionic fields gives the effective action (See Ref. [3]). To avoid critical points sharing thimble structures we complexify G . Taking $G = a + ib$ and $m = 0$, we can find the real and imaginary parts of the action. They are given

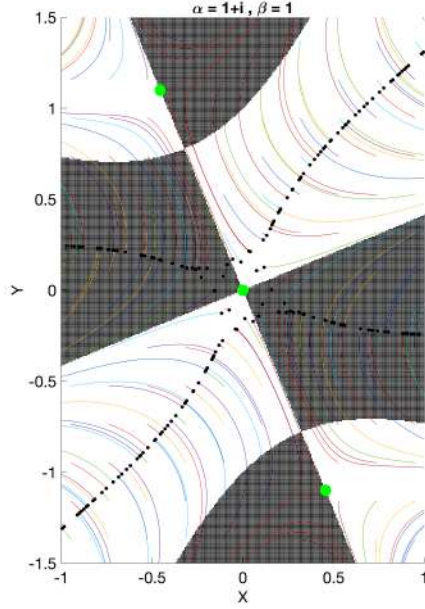


Figure 2: Gradient plots for $\alpha = 1+i$ and $\beta = 1$ with M_0 ODE set. The critical points are the green blobs. The black dots represent the end points of the trajectories.

as

$$\begin{aligned} \text{Re}S[x, y] &= \frac{(x^2 - y^2)a + 2xyb}{a^2 + b^2} \\ &\quad - \frac{1}{2} \ln \left((p^2 + x^2 - y^2)^2 + 4x^2y^2 \right), \end{aligned} \quad (17)$$

$$\text{Im}S[x, y] = \frac{2xya - b(x^2 - y^2)}{a^2 + b^2} - \arctan \left(\frac{2xy}{p^2 + x^2 - y^2} \right). \quad (18)$$

Performing a similar analysis we find the critical points to be $\sigma_c = 0, \pm\sqrt{G - p^2}$. In all figures we show the thimbles and anti-thimbles for $G = 0.7e^{0.1i}$ and $p = 1$. From our understanding we find that the above contours cannot be solved for analytically. The contours were produced in Ref. [3]. Simple contour plots as suggested by the reference fail to reproduce the paper's results.

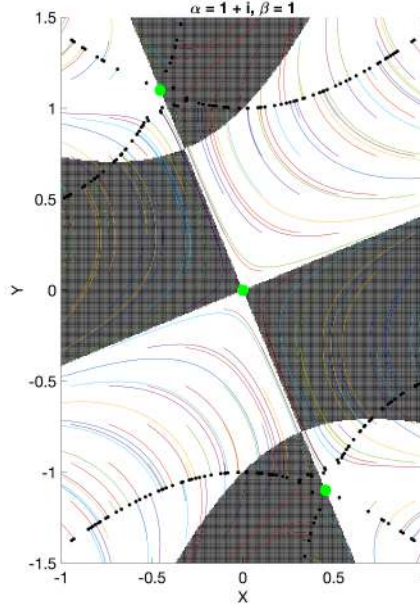


Figure 3: Gradient plots for $\alpha = 1 + i$ and $\beta = 1$ with $M_{\pm}$ ODE set. The critical points are the green blobs. The black dots represent end points of the trajectories.

We suspect that plots from the paper were obtained from fitting curves through the available contour points. This is also the reason for trying out the TC approach on the logarithmic action to see if we acquire better plots resembling the plots in the paper.

In view of the results we still have difficulty in producing the exact same plot, though the overall outline is visible. The best that can be done is collecting piecewise regions of the plots and fitting curves to them. See Figs. 6, 7, 8, 9, and 10 for all the plots.

3 Demerits of the Algorithm

We found the following demerits of the TC algorithm.

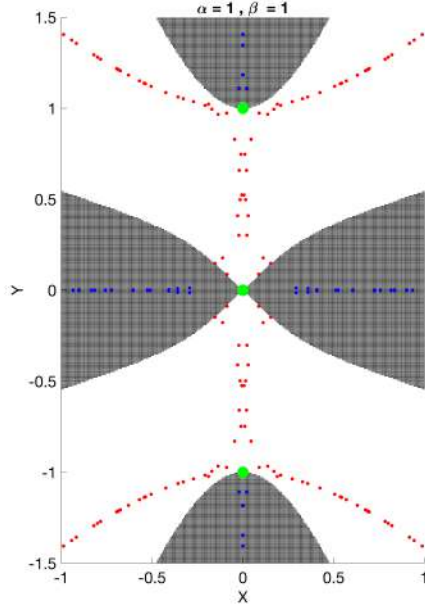


Figure 4: Shaded regions depict the regions of stability, $\text{Re}(S[\phi]) \geq 0$. The red dotted lines are part of thimbles and the blue dotted lines those of anti-thimbles. The green blobs represent critical points. We used $\alpha = 1$ and $\beta = 1$.

- The algorithm is very dependent on finding the the gradient equations analytically. Simply because these equations are highly cumbersome to evaluate by hand, symbolic code will have to be used. These can generally result in some numerical coefficient errors. This is also what we suspect could be leading to incomplete plots for the GN model action.
- The differential equations may play out to be stiff and hence relying on simple Runge-Kutta schemes may prove to be error prone.
- Since the nature of the gradient flow changes in the different regions, filtering out points becomes very difficult when ghosts, thimbles and anti-thimbles are involved in a one region. This is a common problem when dealing with the GN model.

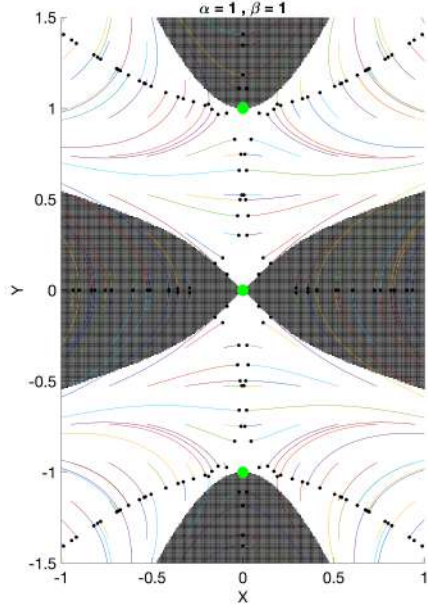


Figure 5: Gradient plots for $\alpha = 1$ and $\beta = 1$. The critical points are the green blobs. The black dots represent end points of the trajectories.

- Also filtering schemes just based on the sign of the $\text{Re}S[\phi]$ will not work because there can be regions of thimbles that lie in $\text{Re}S[\phi] < 0$.
- At best the method can provide a point plot and backbone of thimble structures. Thus in cases where phase transitions in thimbles are to be witnessed, TC method can aid in this development for complicated actions.

4 Algorithm Description with Code Snippets

The outline of the algorithm is provided below.

- Symbolically compute the gradient flow equations. Vectorize these in the MATLAB sense, meaning make the gradient equation take in a vector of initial conditions.

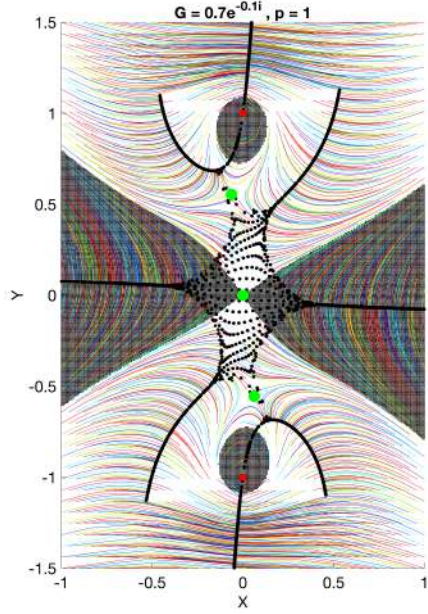


Figure 6: Gradient plots for M_0 ODE set. The critical points are the green blobs. The red squares are the locations where the action diverges. The black dots represent the end points of the trajectories.

- Construct a grid of initial conditions. (Grids can be made with grid centers close to the critical point.)
- Solve for the trajectories with user given time duration
- Collect the end time (x, y) points of the trajectories
- Process these points depending upon the sign of $\text{Re}(S[x, y])$. If negative collect in the anti-thimble array and if not in a thimble array.
- Plot point plots for the thimble array in blue and anti-thimble array in red.
- For interpolation of the structures use a distance function between consecutive points. Segregate ones which are quite distant from a curve forming collection

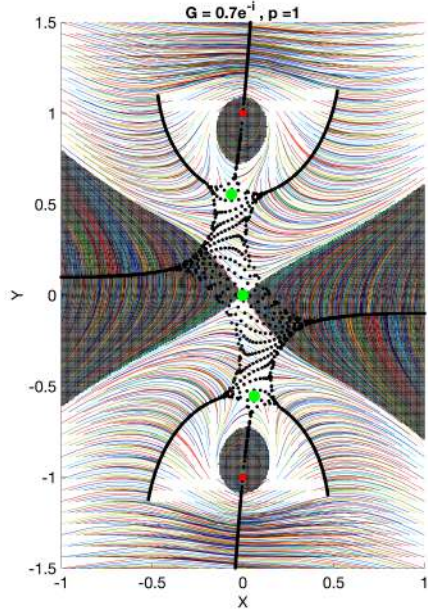


Figure 7: Gradient plots for $M_{\pm}$ ODE set. The critical points are the green blobs. The red squares are the locations where the action diverges. The black dots represent the end points of the trajectories.

of points using a user defined error margin.

We provide the base code structure below.

Initialize the matrix

```
function mat = matrix_initial(grid_size, crit_x, crit_y)
mat=zeros(50^2,2);
grid_x = linspace(-0.5+crit_x,0.5+crit_x,grid_size);
grid_y = linspace(-0.3+crit_y,0.3+crit_y,grid_size);
ind = 1;
for i=1:max(size(grid_x))
    for j=1:max(size(grid_y))
        mat(ind,1)=grid_x(i);
        mat(ind,2)=grid_y(j);
        ind=ind+1;
    end
end
end
```

Create the function handle for ODE solving

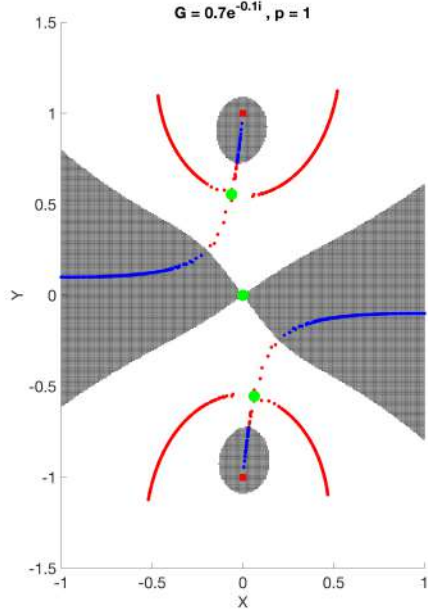


Figure 8: Filtered plots for $M_{\pm}$ ODE set. The critical points are the green blobs. Red and blue dots represent anti-thimbles and thimbles, respectively. A combination of them not passing through critical points leads to ghosts. The red squares are the locations where the action diverges.

```
function dy = ODEfun(t,y)
    Np = numel(y)/2;
    dy = zeros(2*Np,1);
    dy(1:Np,1) = $ \pdv{M}{x}$;
    dy(Np+1:2*Np,1) = $ \pdv{M}{y}$;
end
```

Use `ode45` in MATLAB to perform all calculations.

We provide the plotting code for the zero-dimensional fermionic system.

```
G = 0.7*exp(-0.1*i);
p=1;
x = linspace(-2,2,500);
[L,K] = meshgrid(x);
I = (real((L+i*K).^2./(G - log(p^2 + (L+i*K).^2)) >= 0);
scatter(L(I),K(I),'.','MarkerEdgeColor',[0.25, 0.25, 0.25], 'MarkerFaceColor',[0.25, 0.25, 0.25]);
xlabel('X'); ylabel('Y');
hold on
x2 = real(sqrt(G-p^2));
y2 = imag(sqrt(G-p^2));
Imsc = imag((x2+i*y2).^2./(G - log(p^2 + (x2+i*y2).^2));
```

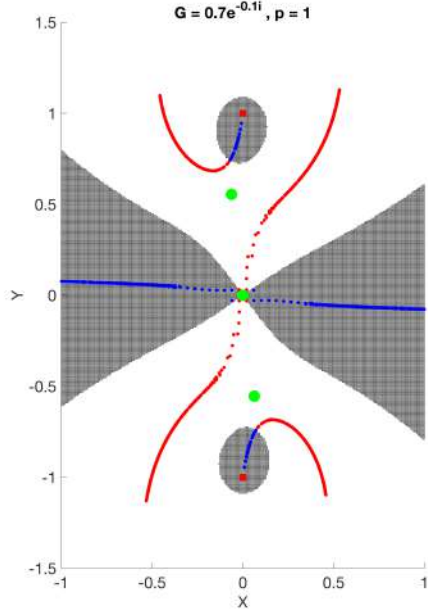


Figure 9: Filtered plots for M_0 ODE set. The critical points are the green blobs. Red and blue dots represent anti thimbles and thimbles, respectively. A combination of them not passing through critical points leads to ghosts. The red squares are the locations where the action diverges.

```

for j = 1:2500
    z = xt_b(end,j) + i * yt_b(end,j);
    S = (z).^2./(G) - log(p^2 + (z).^2);
    if imag(S) <= Imsc + 2*10^-3 && imag(S) >= Imsc - 2*10^-3
        if real(S) < 0
            plot(xt_b(end,j), yt_b(end,j), 'r.', 'MarkerSize', 15, 'color', 'red');
            hold on
        else
            plot(xt_b(end,j), yt_b(end,j), 'b.', 'MarkerSize', 15, 'color', 'blue');
            hold on
        end
    end
    z = x1(end,j) + i * y1(end,j);
    S = (z).^2./(G) - log(p^2 + (z).^2);
    if imag(S) <= 2*10^-3 && imag(S) >= -2*10^-3
        if real(S) < 0
            plot(x1(end,j), y1(end,j), 'r.', 'MarkerSize', 15, 'color', 'red');
            hold on
        else
            plot(x1(end,j), y1(end,j), 'b.', 'MarkerSize', 15, 'color', 'blue');
            hold on
        end
    end
end
end

```

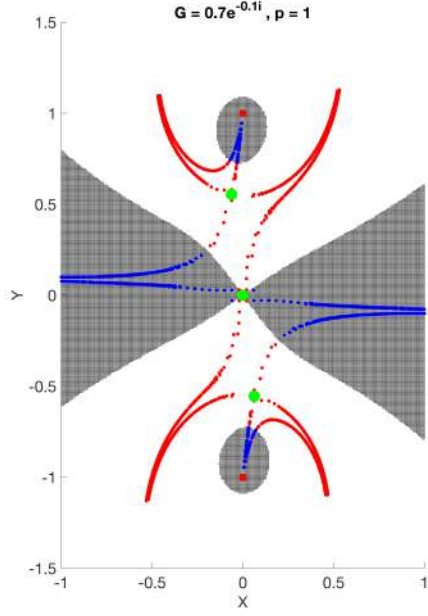



Figure 10: Filtered plots for $M_{\pm}$ and M_0 ODE set. The critical points are the green blobs. Red and blue dots represent anti thimbles and thimbles, respectively. A combination of them not passing through critical points leads to ghosts. The red squares are the locations where the action diverges.

```

end
plot(0,0,'.', 'MarkerSize',40, 'color',[0.8500, 0.3250, 0.0980]);
hold on
plot(real(sqrt(G-p^2)),imag(sqrt(G-p^2)),'.', 'MarkerSize',40, 'color',[0.8500, 0.3250, 0.0980]);
hold on
plot(-real(sqrt(G-p^2)), -imag(sqrt(G-p^2)),'.', 'MarkerSize',40, 'color',[0.8500, 0.3250, 0.0980]);
hold on
plot(0,1,'.', 'MarkerSize',40, 'color','r');
hold on
plot(0,-1,'.', 'MarkerSize',40, 'color','r');
hold on
view(2)
axis([-1 1 -1.5 1.5 ])
daspect([1 1 1])
title('Gradient lines for Thimble Cooling approach GN-Like Model')
xlabel('x')
ylabel('y')

```

5 Supersymmetric Setup

The partition function takes the form

$$Z = \int \mathcal{D}\phi \mathcal{D}\psi \mathcal{D}\bar{\psi} e^{-S[\phi, \psi, \bar{\psi}]}, \quad (19)$$

where

$$S[\phi, \psi, \bar{\psi}] = \int_0^\beta d\tau \left[\frac{\mathcal{B}^2}{2} + i\mathcal{B} \left(\dot{\phi} + W'(\phi) \right) + \bar{\psi} \left(\frac{\partial}{\partial \tau} + \frac{\partial^2 W}{\partial \phi^2} \right) \psi \right]. \quad (20)$$

Here $\mathcal{B}$ is an auxiliary field. The above action defines a $0+1$ dimensional set up. To acquire the 0-dimensional set up we will have to set all the partial derivatives with respect to τ to 0. Thus the effective action in 0 dimensions is

$$S[\phi, \psi, \bar{\psi}] = \frac{\mathcal{B}^2}{2} + i\mathcal{B}W'(\phi) + \bar{\psi} \frac{\partial^2 W}{\partial \phi^2} \psi, \quad (21)$$

where $W(\phi)$ is called the superpotential. Further, the following action can be simplified by integrating out the auxiliary field using its equation of motion $\mathcal{B} = -iW'$. The resulting action is

$$S[\phi, \psi, \bar{\psi}] = \frac{W'^2}{2} + \bar{\psi} W'' \psi. \quad (22)$$

Integrating over the fermionic fields we get

$$S[\phi] = \frac{(W'(\phi))^2}{2} - \log(W''(\phi)). \quad (23)$$

For different $W'(\phi)$ we could try to procure for Lefschetz thimble structures. (Ref. [1] and [2].)

6 Double-Well Potential

The action we are interested has the form

$$S[\phi] = -\ln(W''(\phi)) + \frac{(W'(\phi))^2}{2}. \quad (24)$$

For the double-well potential case we have

$$W'(\phi) = e^{i\alpha} g(\phi^2 + \mu^2), \quad (25)$$

$$W''(\phi) = 2e^{i\alpha} g\phi. \quad (26)$$

Thus the action is

$$S[\phi] = e^{2i\alpha} \frac{g^2}{2} (\phi^2 + \mu^2)^2 - \ln(2e^{i\alpha} g\phi). \quad (27)$$

6.1 Real and Imaginary Parts

Setting $\phi \rightarrow x + iy$, we find that

$$\begin{aligned} \text{Im}(S[\phi]) &= -\alpha - \arctan\left(\frac{x}{y}\right) + \frac{1}{2} \left[\cos(2\alpha) 4xy(x^2 - y^2 + \mu^2) \right. \\ &\quad \left. + \sin(2\alpha) [(x^2 - y^2 + \mu^2)^2 - 4x^2y^2] \right], \end{aligned} \quad (28)$$

and

$$\begin{aligned} \text{Re}(S[\phi]) &= -\frac{\ln(4(x^2 + y^2))}{2} + \frac{1}{2} \left(\cos(2\alpha) [(x^2 - y^2 + \mu^2)^2 - 4x^2y^2] \right. \\ &\quad \left. - \sin(2\alpha) 4xy(x^2 - y^2 + \mu^2) \right). \end{aligned} \quad (29)$$

6.2 Critical Points

There are four critical points as we find the first derivative reduces to a quartic

$$\phi^2 = \mp \frac{\mu^2}{2} \sqrt{\mu^4 + 2e^{-i2\alpha}}. \quad (30)$$

Hessian was calculated at each of these points and was found to be non-zero hence degenerate.

6.3 Parameterizations

The parameterizations $\frac{x}{y} \rightarrow t$ and $\frac{y}{x} \rightarrow t$ and several piecewise t sets will provide us with four $x(t)$ and $y(t)$ for two critical points. Implying that two critical points share the same set of curves on different domains.

The curves are obtained by solving for x in the following equation

$$\begin{aligned} &(\sin(2\alpha) [(1 - t^2)^2 - 4t^2] + \cos(2\alpha) 4t(1 - t^2)) x^4 \\ &+ (\sin(2\alpha) (2\mu^2(1 - t^2)) + \cos(2\alpha) \mu^2 4t) x^2 \\ &+ (2\text{Im}(S(z_c)) - 2\alpha - 2\arctan(t) + \mu^4) = 0. \end{aligned} \quad (31)$$

Similarly for y this can be done. Then reduced quadratic can be solved to acquire thimble structures.

6.4 Cases Considered

We chose $\alpha = 0$, $g = 1$ and $\mu = 2$ and also $\alpha = 0.01$.

The results were got and plotted. There is also a figure that depicts the piecewise plots. See Fig. 11 for all the plots.

Table 1 below goes in sync with the plot from Fig. 11 (c). The first column represents the curve number corresponding to the plot. The numbers are marked on the plot. Each row of the table corresponds to one labelled curve on the plot. The second column gives the parametrization used. Once this is fixed it leads to Eq. (31) or the similar equation where y is the quartic variable to solve for. The quartic equations can be solve like quadratic ones therefore the solution for x or y will have the form $\pm\sqrt{A(t)} \pm \sqrt{B(t)}$. You can see that $\pm$ appears twice, so we can label these solutions as $++$, $--$, $+-$ and $-+$. Where the first sign appears before the first root when read left to right. If you notice in Eq. (31) there is the $\text{Im}(S(z_c))$ that needs to be inputted, the next column provides you with this input. The last column provides us with the domain of t on which the curve was plotted.

Curve Number	Parametrization Type	Signs before Root	$\text{Im}(S(z_c))$	Numerical Domains
1	$t = \frac{y}{x}$	--	0	$t \in [-6, 6]$
2	$t = \frac{y}{x}$	-+	0	$t \in [-6, 1]$
3	$t = \frac{y}{x}$	-+	0	$t \in [1, 6]$
4	$t = \frac{x}{y}$	++	$\pi/2$	$t \in [0, 1]$
5	$t = \frac{x}{y}$	-+	$\pi/2$	$t \in [0, 1]$
6	$t = \frac{-x}{y}$	--	$-\pi/2$	$t \in [-1, 0]$
7	$t = \frac{-x}{y}$	+-	$-\pi/2$	$t \in [-1, 0]$

Table 1: Table elaborating the piecewise plots in Fig. 11 B.

7 Scarf I Potential

The action we are interested has the form

$$S[\phi] = -\ln(W''(\phi)) + \frac{(W'(\phi))^2}{2}. \quad (32)$$

For the Scarf I case we have

$$W'(\phi) = A \tan(\mu\phi) - B \sec(\mu\phi), \quad (33)$$

provided $\mu\phi \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $A > B \geq 0$ and $\mu > 0$. For simplicity we set $B = 0$.

7.1 Critical Points

We now find the critical points for the function

$$S[\phi] = \frac{A^2}{2} \tan(\mu\phi) + 2 \ln(A\mu \cos(\mu\phi)). \quad (34)$$

We have

$$\frac{\partial S}{\partial \phi} = [2 - A^2 \sec^2(\mu\phi)] \mu \tan(\mu\phi) = 0. \quad (35)$$

This gives

$$\tan(\mu\phi) = 0, \quad (36)$$

or

$$2 - A^2 \sec^2(\mu\phi) = 0. \quad (37)$$

These set of equations gives us the critical points and they are

$$\phi = 0, \quad (38)$$

$$\phi_+ = \frac{1}{\mu} \sec^{-1} \left(\sqrt{\frac{2}{A^2}} \right), \quad (39)$$

$$\phi_- = \frac{1}{\mu} \sec^{-1} \left(-\sqrt{\frac{2}{A^2}} \right). \quad (40)$$

We can see that $A > \sqrt{2}$ the $\{\phi_+, \phi_-\} \in \mathbf{R}$ and for $A < \sqrt{2}$ the set is purely imaginary. At $A = \sqrt{2}$ there is a collapse and $\phi = 0$ is the only critical point and it turns out to be degenerate. Hence we will be restricted to either seeing the critical points on the y -axis or purely the x -axis, with the common critical point $(0, 0) \forall A$. The Hessian of the $\text{Re}(S(\phi))$ and $\frac{\partial^2 S}{\partial \phi^2}$ are non zero at all critical points defined for $A \in (-\infty, \sqrt{2}) \cup (\sqrt{2}, 0)$.

7.2 Branch Points and Branch Cuts for $\ln \cos(\phi)$

The principal domain where the $\text{Re}(\cos(\phi)) > 0$ is $(-\frac{\pi}{2}, \frac{\pi}{2})$. The branch points for $\ln \cos(\phi)$ would be $\phi = -\frac{\pi}{2}, \frac{\pi}{2}$.

Due to periodicity and focusing on the interval of interest, we could very well take the branch cuts where the $\text{Re}(\cos(\phi)) < 0$.

Focusing on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ for the $\text{Re}(\phi)$ will not cause any trouble when trying to find the thimble structures. Clarification is still required on this regard to really understand thimble structures for critical points being real.

7.3 Some Thimble and Region Analysis

The real and imaginary parts of the action are

$$\begin{aligned} \text{Re}(S[\phi]) = & -\left(\frac{-A^2}{2} \frac{\sin^2(2x) - \sinh^2(2y)}{(\cos(2x) + \cosh(2y))^2} \right. \\ & \left. + \ln(A) - 2 \ln(\cos^2(x) \cosh^2(y) + \sin^2(x) \sinh^2(y)) \right), \end{aligned} \quad (41)$$

$$\begin{aligned} \text{Im}(S[\phi]) = & -\left(-A^2 \frac{\sin(2x) \sinh(2y)}{(\cos(2x) + \cosh(2y))^2} \right. \\ & \left. - 2 \arctan(-\tan(x) \tanh(y)) \right). \end{aligned} \quad (42)$$

By inspection of where the critical points lie (either on the x -axis or on the y -axis including the origin). We see that for $x = 0$ or $y = 0$ we have $\text{Im}(S[\phi]) = 0$. There for we must look for contours to define the thimble and anti-thimble given by this level set

$$-A^2 \frac{\sin(2x) \sinh(2y)}{(\cos(2x) + \cosh(2y))^2} - 2 \arctan(-\tan(x) \tanh(y)) = 0. \quad (43)$$

We can easily see that $y = 0$ defined in the region $(-\frac{\pi}{2}, \frac{\pi}{2})$ and $x = 0$ satisfy the above level set. It turns out for the case $A \geq \sqrt{2}$ we can easily locate the one of each thimble or anti-thimble for each critical point. For the case $A \geq \sqrt{2}$ we have $x = 0$ as the common thimble for all critical points. The critical point of the origin consists of the anti-thimble defined as $y = 0$ which ends at $x = \pi/2, -\pi/2$, the dual is the $y = 0$ line. Though we would like to suggest the partitioning of the contours $x = 0$ and $y = 0$ lines into thimble regions and anti-thimble regions will sort the confusion for all critical points.

For every critical point in this set up we see common contour $x = 0$ and $y = 0$ lines.

Solving for the thimble for the critical points ϕ_+ and ϕ_- analytically is proving to be quite difficult, due to dealing with inverse tangent as well as hyperbolic trigonometric functions. In reference to the TC algorithm, manual ODE tolerance is failing because of hyperbolic trigonometric functions bring in exponentials.

From contour plots of $Im(S[\phi]) = Im(S[\phi_{\pm}])$ it is suspected that the anti-thimbles at ϕ_+ and ϕ_- are symmetric and form a closed loop in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$. The graphs were produced using MATLAB. See Fig. 12 for all the plots.

7.4 Behavior of the thimbles under change of parameter A

We can see from Fig. 12 that there are 4 different plots for different parameter values of A . In spite of not arriving at all the thimble and anti-thimble structures for all the critical points we can make a positive analysis to understand this model. Important fact we see is that the contours $x = 0$ and $y = 0$ are common to one of the critical points when $A \in \mathbb{R}$. When $A \in \mathbb{C}$ the contours solely contribute to the $(0, 0)$ critical point. From our previous analysis we concluded that the $(0, 0)$ is always a critical point of the action irrespective of the parameter A . Following this we can see that the $x = 0$ and $y = 0$ lines will always be contours irrespective of A . So changing A keeps the contours of $(0, 0)$ and the critical point itself invariant.

But we can say that the break up of regions of these contours into thimbles and anti-thimbles can change as you vary the parameter A . This is noticeable when moving from $A = 1$ to $A = \sqrt{3}$. The critical points which were originally lying on the x -axis shifts to the y -axis. It also looks as though the thimbles structures of $A = 1$ have sort of rotated by 90 to arrive at the structures of $A = \sqrt{3}$. This tells us that just by changing the real parameter A we might be able to look for a phase transition. This also provides us with new directions to look into.

8 Morse Potential

The action we are interested has the form

$$S[\phi] = -\log(W''(\phi)) + \frac{(W'(\phi))^2}{2}. \quad (44)$$

For the Morse function case we have

$$W'(\phi) = A - Be^{-\alpha\phi} \quad \text{and} \quad W''(\phi) = B\alpha e^{-\alpha\phi}. \quad (45)$$

Now plugging these into the action we get

$$S[\phi] = \frac{(A - Be^{-\alpha\phi})^2}{2} + \alpha\phi - \ln(B\alpha). \quad (46)$$

Next step is to find the critical points. After that we move on to complexify $\phi \rightarrow x + iy$. We will also assume that $A, B, \alpha \in \mathbb{R}$.

8.1 Critical Points

We have

$$\frac{\partial S}{\partial \phi} = -B\alpha e^{-\alpha\phi}(A - Be^{-\alpha\phi}) - \alpha = 0. \quad (47)$$

Making the textbook substitution $e^{-\alpha\phi} \rightarrow t$ simplifies to solving a quadratic with roots

$$t_{\pm} = \frac{A \pm \sqrt{A^2 + 4}}{2B}. \quad (48)$$

Hence

$$\phi_{\pm} = -\frac{1}{\alpha} \ln\left(\frac{A \pm \sqrt{A^2 + 4}}{2B}\right). \quad (49)$$

We can clearly see that $t_{\pm}$ would either be positive or negative for any value of A, B and α , for the sake of clarity the roots will be always of opposite sign. Implying that one of the $\phi_{\pm}$ is real and the other complex lying on either $y = \frac{\pi}{\alpha} / -\frac{\pi}{\alpha}$ for some x . We were also able to verify that $\frac{\partial^2 S}{\partial \phi^2} \neq 0$ at the critical points and also the Hessian of $\text{Re}(S[\phi])$ is not zero at these critical points.

We have no doubts that critical points are non-degenerate and Morse theory can be applied to arrive at Lefschetz thimbles for the given action.

8.2 Real and Imaginary Parts of the Action

After calculations we get the following form for the real and imaginary parts of the action

$$\begin{aligned} \text{Re}(S(x, y)) &= -\left(-\frac{A^2}{2} + \ln(B\alpha) - \frac{B^2}{2}e^{-2\alpha x} \cos(2\alpha y) \right. \\ &\quad \left. + AB e^{-\alpha x} \cos(\alpha y) - \alpha x\right), \end{aligned} \quad (50)$$

$$\text{Im}(S(x, y)) = -\left(\frac{B^2}{2}e^{-2\alpha x} \sin(2\alpha y) - AB e^{-\alpha x} \sin(\alpha y) - \alpha y\right). \quad (51)$$

8.3 Analytic Expressions for Level Sets $\text{Im}(S[\phi]) = \text{Im}(S[\phi_{\pm}])$

For the sake convenience we will label the roots as ϕ_{real} and ϕ_{complex} .

Now considering ϕ_{real} we see that $y = \text{Im}(\phi_{\text{real}}) = 0$. Substituting this value of y we find that $\text{Im}(S[\phi_{\text{real}}]) = 0$. For the critical point ϕ_{real} the contours we are interested in come from $\text{Im}(S(x, y)) = 0$.

We could also easily find the contours associated to this by making the classic substitution $e^{-\alpha x} \rightarrow t$ and solving for the resulting quadratic. We also see that certain parameterizations sets give the resulting curves passing through the required critical point.

The general expression, after solving for and then simplification, takes the following form, for a general setup

$$t_{\pm}^{\text{real/complex}} = \frac{A \pm \sqrt{A^2 + 4 \cot(\alpha y)(\alpha y + \text{Im}(S[\phi_{\text{real/complex}}]))}}{2B \cos(\alpha y)}. \quad (52)$$

We can also see how $y \rightarrow 0$, and given the fact that $\text{Im}(S[\phi_{\text{real}}])$ is used, we see that $t_{+}^{\text{real}} \rightarrow \phi_{\text{real}}$. It turns out that $y \in (-\frac{\pi}{2\alpha}, \frac{\pi}{2\alpha})$ captures every details of a contour passing through the critical point. The other contour passing through ϕ_{real} turns out to be $y = 0$ (x -axis).

While dealing with ϕ_{complex} we should note that $y = -\frac{\pi}{\alpha}$. Due to this we have $\text{Im}(S[\phi_{\text{complex}}]) = \pi$ for any x . We see that $y \rightarrow -\frac{\pi}{\alpha} \implies t_{-}^{\text{complex}} \rightarrow \text{Re}(\phi_{\text{complex}})$. There are definitely segments of $y \in (-\frac{3\pi}{2\alpha}, -\frac{\pi}{2\alpha})$ for t_{-}^{complex} that captures the details near the critical point. Interestingly t_{+}^{complex} captures almost all the behavior outside a region of the critical point. Hence this contour has three piecewise contours to complete it. The graphs were produced using MATLAB. We used the parameter values $\alpha = \pm 1$, $A = 1$ and $B = 1$.

8.4 Complexifying $B \rightarrow e^{i\theta}$

The critical points get shifted by an amount $\frac{\theta}{\alpha}$ along the y axis.

$$\phi_{\pm} = -\frac{1}{\alpha} \ln\left(A \pm \sqrt{A^2 + 4}\right) + i\frac{\theta}{\alpha}. \quad (53)$$

The imaginary and real part of the action after some simplification can be written as

$$\text{Im}(S[x, y]) = -\left(\theta - \alpha y - \frac{e^{-2\alpha x} \sin(2\theta - 2\alpha y)}{2} + Ae^{-\alpha x} \sin(\theta - y\alpha)\right), \quad (54)$$

$$\begin{aligned} \text{Re}(S[x, y]) = & -\left(-\frac{A^2}{2} + \ln(\alpha) - \alpha x - \frac{e^{-2\alpha x} \cos(2\theta - 2\alpha y)}{2}\right. \\ & \left.+ Ae^{-\alpha x} \sin(\theta - \alpha y)\right). \end{aligned} \quad (55)$$

Thus the substitution $\theta - \alpha y \rightarrow \alpha' y'$ in the main parametric equations will give us the structures of interest. For $\theta = 1$ and keeping all other parameters same as above we have generated the plots. They are given in Fig. 13. Even without plots we can conclude that the shift in the y - coordinate just shifts thimble structures. So the overall characteristics are the same.

9 Closing Remarks

We have been able to produce the thimble structures for the double-well potential analytically. Complexification of the double-well parameters also yields solvable structures. They seem to have a characteristic one-thimble contribution. The half $y = 0$ axis is the thimble and this is the integration over the half-real line, though the original integration cycle, without the complexification, is over the entire real line. Computing these integrations can probably give us a framework for answering the question on using thimbles to understand SUSY breaking. The piecewise plots also tell us how complicated the relationship is between the contour structure and the parametrization. We would need to find an effective way to break up the relevant domains of the parametrizing variable. This is important as the thimble analysis is not only interested in the neighborhood of the critical point but also in the behavior of the anti-thimble with real line.

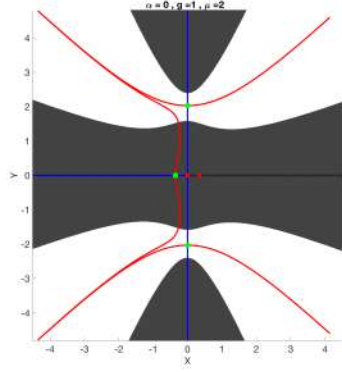
For the case of the Scarf I potential, the imaginary part of the action was too complicated to be solved using analytical methods. Even contour plots exhibited numerical inconsistencies giving us with fractal plots and without any clear boundary plots. Though $x = 0$ and $y = 0$ lines are the only observed contours, their construction does not help us in classifying them as thimble or anti-thimble, in spite of the function respecting the Morse conditions. We suspect that a careful labelling of the branch cuts might help us sort out this confusion. With a change in the parameter A in this model we could analyze the nature of the critical points and regions of convergence. We could also understand the dynamics of the thimble structures for the common critical point at the origin. The Scarf I potential could be a good model to study phase transitions as a simple parameter change leads to a very drastic change in positioning of the critical points.

The Morse potential was the simplest to deal with. Elementary school grade mathematics was all that was needed to specify the thimbles. Also, the thimble analysis resonates with actual calculation of the partition function. In resemblance to the double well potential, the $y = 0$ line is a thimble and also the only contributing one in the calculation of the partition function. Added to this, the integration cycle matches with the theoretical one for all parameters being real. The complexification

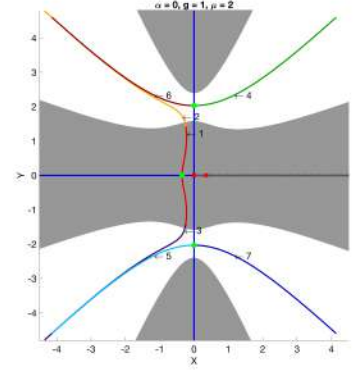
of B turns out to be non-trivial in this regard.

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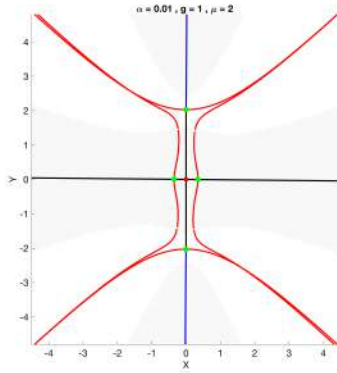
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(a)

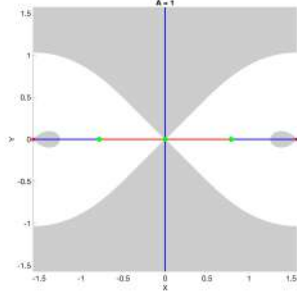


(b)

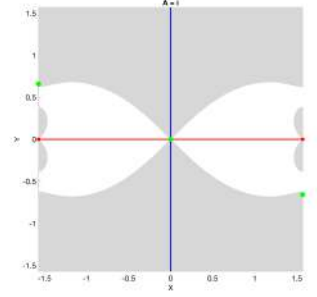


(c)

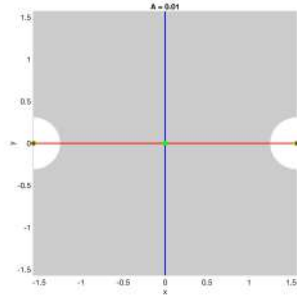
Figure 11: (a) The thimbles and anti-thimbles are represented by blue and red colored curves. The shaded regions represent the region of convergence, $\text{Re}(S[\phi]) \geq 0$ for $\alpha = 0$, $g = 1$ and $\mu = 2$. The crossed line is the branch cut. Critical points are the green blobs. The second red square after the origin is a critical point lying on the branch cut. (b) Piecewise plots of the thimbles. (c) Red: anti-thimble and blue: thimble. Shaded region represents the region of convergence, $\text{Re}(S[\phi]) \geq 0$ for $\alpha = 0.01$, $g = 1$ and $\mu = 2$. Critical points are the green blobs. Black line represents possible thimbles that have not been derived yet from the analytic expressions.



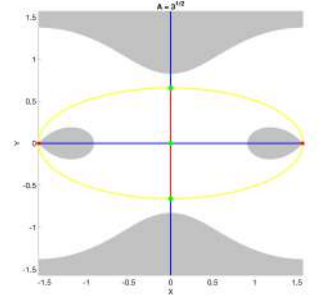
(a) The shaded region represents the region of convergence, $\text{Re}(S[\phi]) \geq 0$ for $A = 1$.



(b) The shaded region represents the region of convergence, $\text{Re}(S[\phi]) \geq 0$ for $A = i$.



(c) The shaded region represents the region of convergence, $\text{Re}(S[\phi]) \geq 0$ for $A = 0.01$.



(d) The shaded region represents the region of convergence, $\text{Re}(S[\phi]) \geq 0$ for $A = \sqrt{3}$.

Figure 12: Plots of the thimbles and their duals for Scarf I case for different values of A . The red squares are $(\frac{\pi}{2}, 0)$ and $(-\frac{\pi}{2}, 0)$ (branch cuts start from these points). The action diverges at these points. Green blobs depict the critical points. Blue lines are the thimbles and the red ones are the duals. The yellow line in (d) is a predicted anti-thimble from contour plotting.

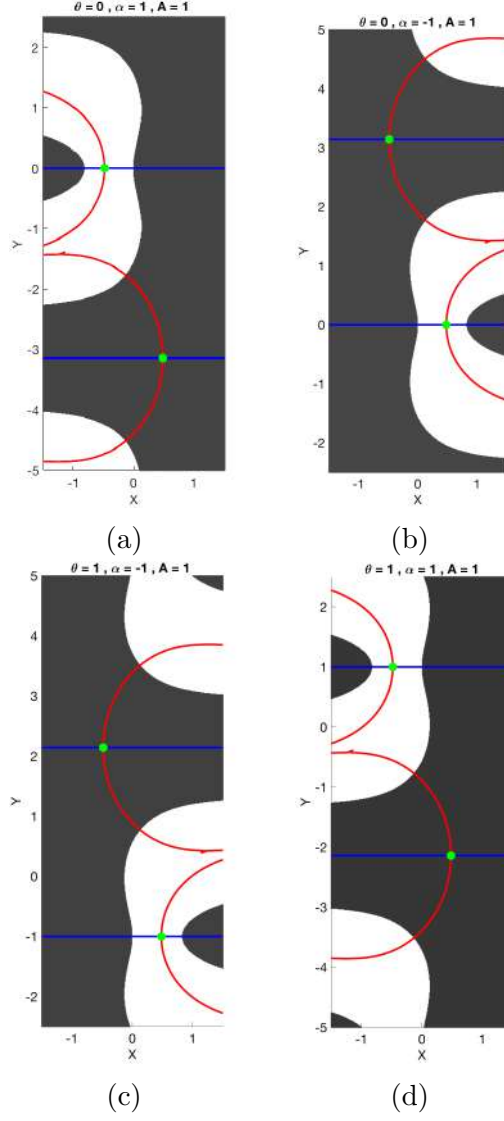


Figure 13: In all the figures the shaded regions represent the regions of stability and green blobs represent the critical points. The parameters are (a) $\theta = 0$, $\alpha = 1$ and $A = 1$; (b) $\theta = 0$, $\alpha = -1$ and $A = 1$; (c) $\theta = 1$, $\alpha = -1$ and $A = 1$; and (d) $\theta = 1$, $\alpha = 1$ and $A = 1$.